

Amorphous crystals

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Quasi-brittle yielding is a salient feature of well-annealed glassy materials. Here we show that almost the same phenomenon characterizes plastic yielding in model 2D perfect crystals. In such systems the elastic stage of deformation terminates with a nonlinear elastic instability resulting in massive nucleation of dislocations. As we show this system spanning (first) yield type event effectively converts the atomic configuration from crystalline to a glassy. The subsequent deformation closely resembles quasi-brittle yielding of the ensuing pseudo-amorphous system. To corroborate these claims we present a detailed study of dislocation dynamics in a prototypical square crystal subjected to quasistatic loading. Our main tool is a novel mesoscopic tensorial model (MTM) of crystal plasticity representing a conceptual trade-off between continuum and atomic descriptions. In addition to overall strain response, the MTM approach allows one to capture intermittent statistics of dislocational avalanches. In close parallel to what is known about plasticity of glassy systems, here we also discover that both pre- and post- (second) yield avalanches in emerging ‘amorphous crystal’ exhibit power law statistics with matching exponents indicating marginal stability of the associated defect configurations.

I. INTRODUCTION

Despite intense focus on the subject in the last years, the phenomenon of *plastic yield* in crystalline solids remains rather mysterious [1–10]. It can be characterized as a more or less abrupt loss of elastic rigidity and it also marks a transition from largely reversible to strongly irreversible deformation. For a crystal subjected to quasi-static loading in a hard device an intriguing related question is why the memory of local atomic neighborhoods, ensuring elastic or quasi-elastic response, suddenly deteriorates allowing the system to abruptly sever a link with its deformation history. An even more challenging question concerns the nature of post-yield plastic flow which often exhibits intermittent fluctuations and shows scale free spatial organization. The goal of the present paper is to address these questions in a narrow but fundamental setting of quasi-brittle plastic yielding of pristine (perfect, defect free) crystals [11–17]. The peculiarity of this particular setting is related to the fact that, rather surprisingly, very similar quasi-brittle plastic yielding is also a salient feature of some glassy materials which lack any crystal symmetry and are structurally indistinguishable from liquids [18–27].

To build a conceptual bridge between plasticity of crystals and glasses, we show that plastic yielding in stylized 2D pristine crystals can be represented as a two-stage process. During the first stage of the yielding, massive dislocation nucleation results in the formation of what is effectively a glassy dislocational configuration, see also [11, 28, 29]. The ensued plastic flow can be viewed as largely ‘caged’ which implies non-collective spatially localized dislocation rearrangements. The second stage of the yielding interrupts this flow abruptly through a system size, catastrophic dislocational avalanche. It is this stage that is very similar to quasi-brittle yielding in amorphous plasticity as it likewise reveals itself through the formation of system-size shear bands. In both crystals and glasses the subsequent plastic flow is stationary with stress reaching a characteristic plateau. It remains to be said that the discussed behavior concerns model crystals with suppressed hardening. More specifically, to corroborate our claims we present a detailed study of dislocation dynamics in a prototypical square crystal subjected to non-generic quasistatic loading by shear strain applied along one of two the slip directions. Our main tool is a novel mesoscopic tensorial model (MTM) of crystal plasticity representing a conceptual trade-off between continuum and atomic descriptions [13–17, 30–32].

In view of the implied crystal-glass analogy, we begin by mentioning some facts about quasi-brittle plastic yield in glassy amorphous systems. It is known that under quasi-static strain control, such systems exhibit initially an almost reversible nominally-elastic response even though the overall stress increase is occasionally interrupted by small stress drops revealing highly localized quasi-plastic rearrangements. At a critical strain such response gives way to fully global plastic deformation leading to stress saturation. In this stationary regime, short elastic stages coexist with stress drops caused by localized plastic shear transformations which occasionally also induce long-range system-spanning rearrangements. The transition to this constant-stress regime, which is the one usually interpreted as plastic yielding, can be of two types. In ductile yielding we observe a continuous crossover towards a stress plateau with only small stress overshoot which can be stretched over a large strain range. In contrast, quasi-brittle yielding involves catastrophic failure through the sudden formation of system-spanning localized shear bands. It was shown that ductile yielding takes place in rapidly quenched glasses with higher degree of disorder while quasi-brittle yielding is characteristic for more structured highly annealed glass samples. In the case of glasses, such bi-modal yielding behavior was successfully reproduced numerically using prototypical elasto-plastic models in which individual linear

elastic elements could yield above some critical stress triggering similar instabilities in other elements [33–37]. Some theoretical approaches have also been advanced to support these empirical findings [38–41].

As we have already mentioned, the main goal of the present paper is to extend the obtained understanding of quasi-brittle yielding from amorphous to crystal plasticity. The two phenomena are sometimes perceived as fundamentally different in view of the observation that dislocations, the main carriers of plastic deformations in crystals, are absent in amorphous solids. However, the elementary mechanical events behind the propagation of individual dislocations and behind the local advance of amorphous plasticity by shear transformation zones, are basically the same and can be in both cases described as triggering of Eshelby-type shear eigenstrains [42–44]. Moreover, behind both amorphous and crystal plasticity is the evolution of elastic incompatibility which can be characterized in both cases by elastic curvature induced by the associated metric tensor [45–49].

In building a connection between crystalline and amorphous plasticity, the first challenge is to choose a relevant modeling framework which would describe adequately crystal dislocations while being still compatible with the modeling language of amorphous plasticity. It is clear that engineering crystal plasticity, which is formulated within classical continuum framework and relies on macroscale phenomenological constitutive relations, is not appropriate for this purpose since even dislocation cores there are not resolved. Numerous computational alternatives to continuum plasticity have been developed ranging from molecular dynamics (MD) [50–53] and similarly microscopic phase-field crystal method (PFC) [54–56] to the more coarse-grained phase field dislocation dynamics (PFDD), [57–60], discrete dislocation dynamics (DDD), [61–67] and, continuum dislocation dynamics (CDD) [68–71]. The multi-scale quasi-continuum model (QC) [72–75], attempted to bridge all the scales while fully comprehensibly accounting for each of them. A detailed discussions of strengths and shortcomings of all these approaches can be found in [13, 14, 16].

They were all successful in addressing particular time and length scales, with higher-scale models (more coarse-grained) relying on input from lower-scale models (less coarse-grained). However, the ability of the existing approaches to capture the correct statistics of plastic fluctuations, while also accounting for large lattice rotations and adequately representing the crystallographic nature of lattice-invariant shears, remains rather limited [76, 77].

To deal with these important aspects of crystal plasticity we rely in this paper on a more recent Landau-inspired MTM approach representing a conceptual trade-off between continuum and atomic descriptions. It can deal adequately with dislocation cores while operating with engineering concepts of stress and strain [13–17, 30–32, 78–87].

It can be viewed as an extension of the local QC model or as a tensorial version of the PFDD, accounting geometrically adequately for both large strains and large rotations. MTM describes faithfully short-range interactions between dislocations and can deal with anisotropic nonlinear elasticity. Moreover, it resolves in continuum setting the full crystallographic symmetry, including lattice-invariant shears [88–92].

Behind the emergence of the continuum stresses and strains in MTM lies the spatial averaging over a mesoscopic length scale: the latter characterizes the size of a maximal cluster of atoms which can be assumed to deform homogeneously. The MTM belongs to the class of mesoscopic elasto-plastic models used in the studies of amorphous plasticity, but instead of using conventional Eshelby elastic propagator, it deals directly with geometrically and physically nonlinear elasticity of inhomogeneous solid bodies and instead of simplified description of plastic slip, it accounts adequately for crystallographically specific lattice invariant shears. The resulting approach has the advantage over DDD of describing without any special phenomenological assumptions, the topological changes in dislocation configurations such as nucleation, annihilation and interaction with obstacles, all in a self-consistent manner and only relying on the assumptions about the inter-atomic potential. Most importantly, it allows one to resolve scale-free dislocational avalanches, responsible for intermittent fluctuations accompanying steady plastic flows in low symmetry crystals. We mention that various other MTM-related approaches have been also explored in the literature [93–102].

In our study of the phenomenon of quasi-brittle yielding of pristine model crystals, we use the most simple but still non-trivial setting. Specifically, to simulate athermal quasistatic loading of perfect square crystals in simple shear, we use the 2D MTM approach, which is thoroughly reviewed to make the presentation fully self contained. As we have already mentioned, our main finding is the two stage structure of plastic yielding. The first stage is a nonlinear elastic instability of a homogeneous crystal which involves dramatic proliferation of dislocations and effectively transforms an ideal crystal into a quasi-amorphous solid. The subsequent deformation of such 'primed' crystal is strongly reminiscent of quasi-brittle plastic yield of a well annealed glass. It involves a 'microplasticity' stage which ends abruptly with a formation of system spanning grain boundaries. The corresponding post-(second)yield plastic deformation closely resembles amorphous plastic flow which is sometimes interpreted as a regime of extended criticality. It is marked by the formation of a stress plateau and involves intermittent avalanche type rearrangements of dislocational patterns which can be both local and global. In parallel, to what is known about plasticity of glassy systems, here we also discover that both pre- and post-(second) yield avalanches exhibit power law statistics with matching exponents indicative of marginal stability.

The rest of the paper is organized as follows. In Section 2 we review the main concepts of the MTM approach. The macroscopic results of our numerical experiments are summarized in Section 3. Statistical analysis of fluctuations is presented in Section 4. The last Section 5 contains our conclusions and indicates some open problems.

II. THE MODEL

In overview, the MTM is a geometrically nonlinear theory of Landau-type where the role of the order parameter is played by the metric tensor (characterizing local deformation). In the corresponding 2D continuum theory the elastic energy density is assumed to respect the full $GL(2, \mathbb{Z})$ symmetry of a Bravais lattices represented by 3×3 invertible matrices $\mathbf{m}$ with integer entries and unit determinants. This implies that such energy density is a multi-well function of tensorial parameter with the bottoms of the energy wells corresponding to lattice invariant deformations. In the numerical implementation of the MTM, the space of admissible deformations is reduced to compatible piece-wise-affine mappings. Therefore, the outcome of the numerical energy minimization depends on the value of the internal length representing the size of the mesoscopic finite element which brings a cut-off scale into the theory. Given the large magnitude of the associated transformation strains, different phases end up being localized at the scale of mesoscopic elements, and the domain boundaries appear macroscopically as linear defects representing mesoscopic dislocations.

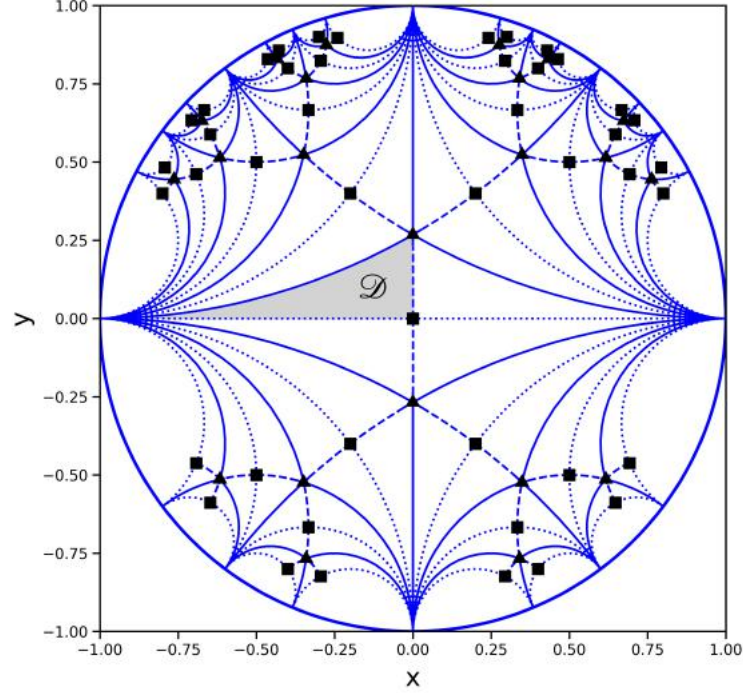


FIG. 1. Tesselation of the metric tensor configurational space into fundamental periodicity domains using Poincaré disk coordinates. The tessellation consists of $GL(2, \mathbb{Z})$ copies of the fundamental domain highlighted in gray. Black squares and triangles denote regions corresponding to high-symmetry square and triangular lattice bases.

A. Elastic energy

In the 2D version of the MTM approach the configurational space of metric tensors is tessellated into an infinite number of equivalent domains, where equivalence is understood in terms of the symmetry transformations induced by the infinite group $GL(2, \mathbb{Z})$. If the energy density is known in one of such tensorial periodicity domains, it can be extended to the whole configurational space. To be more specific, suppose that

$$\mathbf{y} = \mathbf{y}(\mathbf{x}) \quad (1)$$

represents the deformation of a continuum body, where the coordinate $\mathbf{y}$ denotes points in the current configuration and $\mathbf{x}$ refers the reference state. In view of objectivity constraint, the strain energy density of an elastic solid φ can depend on the deformation gradient

$$\mathbf{F} = \nabla \mathbf{y}, \quad (2)$$

only through the Cauchy-Green tensor

$$\mathbf{C} = \mathbf{F}^T \mathbf{F}. \quad (3)$$

To express the global lattice induced symmetry of the ensuing function

$$\varphi = \varphi(\mathbf{C}) \quad (4)$$

which goes beyond the conventional continuum mechanics point group, we need to recall that 2D Bravais lattices can be generated by two linearly independent vectors $\{\mathbf{e}_I\}$, $I = 1, 2$, which form a lattice basis. The two bases $\mathbf{e}_I$ and $\bar{\mathbf{e}}_I$ describe the same lattice if $\mathbf{e}_J = m_{IJ}\bar{\mathbf{e}}_I$ with $m_{IJ} \in \mathbb{Z}$. In this sense, all 2D simple lattices are invariant under the action of a group

$$GL(2, \mathbb{Z}) = \{\mathbf{m}, m_{IJ} \in \mathbb{Z}, \det(\mathbf{m}) = \pm 1\}. \quad (5)$$

Therefore, to preserve a Bravais lattice, the strain energy density of a crystal must respect the symmetry

$$\varphi(\mathbf{C}) = \varphi(\mathbf{m}^T \mathbf{C} \mathbf{m}), \quad (6)$$

where $\mathbf{m}$ belongs to $GL(2, \mathbb{Z})$.

In view of this global symmetry, the surface $\det \mathbf{C} = 1$ (isochoric deformations) is tessellated in the 3D space (C_{11}, C_{22}, C_{12}) into an infinite number of equivalent periodicity domains, see Fig. ???. To better visualize such a tessellation, it is convenient project the infinite surface $\det \mathbf{C} = 1$ stereographically on a (Poincaré) disk with unit radius as illustrated in Fig. ??b. For comparison, we also present in Fig. ??c another orthogonal projection of the same surface on a plane. In both projections the equivalent square lattices are indicated by small black squares while red dots denote equivalent triangular (hexagonal) lattices. As we have already mentioned, it is sufficient to define the energy density only in one minimal periodicity domain, D which is

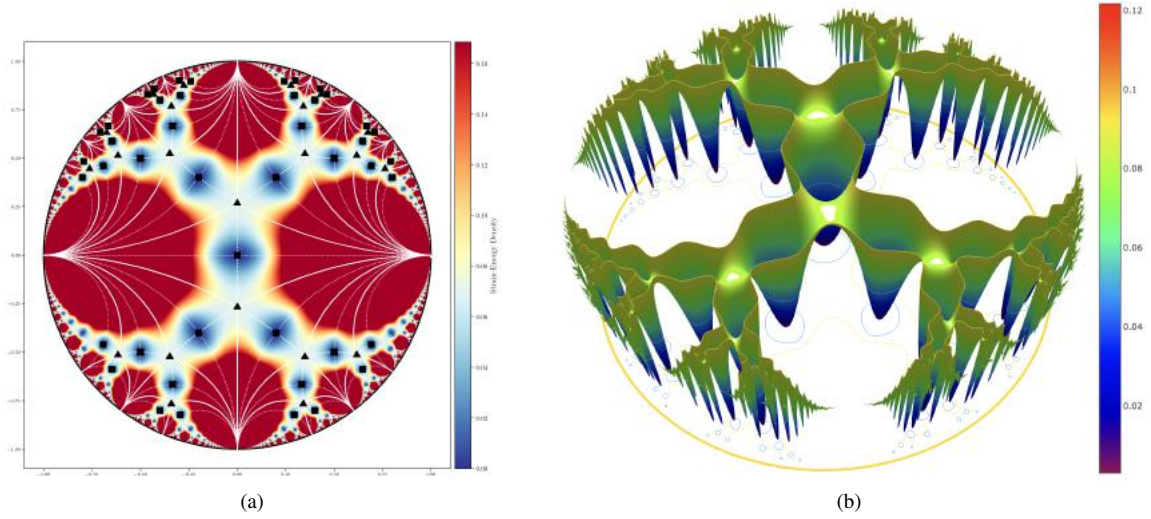


FIG. 2. Strain-energy density visualization in the configurational space of metric tensors using Poincaré disk coordinates. (a) Tessellated representation showing the discrete sampling of the strain-energy density field. (b) Corresponding continuous height plot of the strain-energy landscape over the same coordinate system.

known in the mathematical literature on modular forms as a 'fundamental domain', see shaded domains in our Fig. ??(a,b,c). We shall refer to the restriction of the energy density to the fundamental domain as $\varphi_D(\tilde{\mathbf{C}})$ where $\tilde{\mathbf{C}}$ is the projection of a general $\mathbf{C}$ into this domain. We can write $\tilde{\mathbf{C}} = \mathbf{m}^T \mathbf{C} \mathbf{m}$ where $\mathbf{m}$ is the corresponding mapping of the metric tensor $\mathbf{C}$ into the fundamental domain ensuring that $\varphi(\mathbf{C}) = \varphi_D(\tilde{\mathbf{C}})$.

In our 2D case, illustrated with better resolution in Fig. 2(a), the fundamental domain can be chosen in the form [85, 87, 92, 103]

$$\mathcal{D} = \left\{ C \in \det \mathbf{C} = 1, \quad 0 < C_{11} \leq C_{22}, \quad 0 \leq C_{12} \leq \frac{C_{11}}{2} \right\}. \quad (7)$$

Given a generic metric $\mathbf{C}$, the task of finding a unimodular matrix $\mathbf{m}$ ensuring that $\tilde{\mathbf{C}} \in D$ (and therefore $\varphi(\mathbf{C}) = \varphi_D(\tilde{\mathbf{C}})$) can be formulated in the form of a recursive algorithm known as Lagrange reduction [87, 103]. Specifically, if we define the matrices :

$$\mathbf{m}_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (8)$$

$$\mathbf{m}_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (9)$$

and

$$\mathbf{m}_3 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}. \quad (10)$$

we can formulate the reduction process in a form of an explicit algorithm. Thus, if we start with an assumption that $\mathbf{m} = \mathbb{I}$, we can proceed through the following steps: (i) if $C_{12} < 0$, change sign to C_{12} , $\mathbf{m} \rightarrow \mathbf{m}\mathbf{m}_1$; (ii) if $C_{22} < C_{11}$, swap these two components, $\mathbf{m} \rightarrow \mathbf{m}\mathbf{m}_2$; (iii) if $2C_{12} > C_{11}$, set $C_{12} = C_{12} - C_{11}$, $\mathbf{m} \rightarrow \mathbf{m}\mathbf{m}_3$. As a commentary, we note that the action of the matrix $\mathbf{m}_1$ is a reflection which returns an acute angle between two lattice vectors $\mathbf{e}_i$. The action of the matrix $\mathbf{m}_2$ is also a reflection as it swaps two lattice vectors $\mathbf{e}_i$. Both of these operations belong to the point group and propagate the metric only inside the corresponding 'elastic domain' (Pitteri neighborhood) composed of the four copies of the minimal domain D [92]. Instead, the mapping defined by matrix $\mathbf{m}_3$ brings the metric outside the 'elastic domain' and therefore represents a quantized analog of the macroscopic plastic strain [14].

The single-period Landau potential $\varphi_D(\tilde{\mathbf{C}})$ can be now constructed using the classical Cauchy-Born approach. Suppose that material points in a representative volume Ω undergo an affine deformation such that

$$\mathbf{y}(\mathbf{x}) = \mathbf{x} + \mathbf{u}(\mathbf{x}), \quad (11)$$

and the displacement vector $\mathbf{u}(\mathbf{x})$ is an affine function of $\mathbf{x}$, see Fig. 3. Specifically, if we define the vectors $\mathbf{R} = \mathbf{x} - \mathbf{x}'$ and $\mathbf{r} = \mathbf{y} - \mathbf{y}'$, connecting two atoms in the reference and in the deformed configurations, respectively, we can write $\mathbf{r} = \mathbf{F}\mathbf{R}$ where we introduced the homogeneous deformation gradient $F_{ia} = \partial y_i / \partial x_a = \delta_{ia} + \partial u_i / \partial x_a$. To compute the energy density $\varphi(\mathbf{C})$, we need to account for the elongation or shortening of every atomic bond and then average over the domain Ω . Then, if the interatomic potential is $V(|\mathbf{r}|)$ which is defined within a given cutoff, we can write

$$\varphi(\mathbf{C}) = \frac{1}{2\Omega} \sum_{\mathbf{x}} \sum_{\mathbf{x}' \in \mathcal{N}(\mathbf{x})} V\left(\sqrt{R_a C_{ab} R_b}\right), \quad (12)$$

where $C_{ab} = F_{ia} F_{ib}$ and the internal summations involves all points $\mathbf{x}'$ belonging to the cutoff neighborhood $\mathcal{N}(\mathbf{x})$. After we compute the energy in this way inside the fundamental domain D , we can extend it to the whole configurational space by $GL(2, \mathbb{Z})$ periodicity. The ensuing Landau potential can be expected to become smooth only in the limit when the size of the domain Ω tends to infinity.

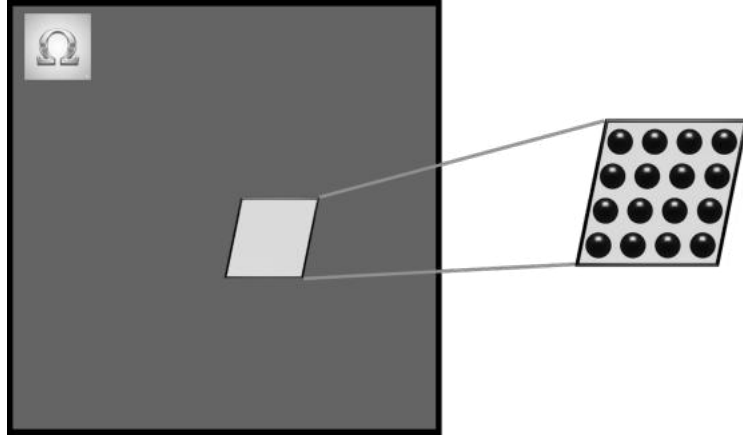


FIG. 3. Schematic representation of the finite element containing a number of atomic units which are assumed to deform homogeneously.

One can also achieve a desired level of smoothness phenomenologically, by representing the energy density inside the domain D using polynomial functions of different degrees. In this work we adopt, for simplicity, such an approach and without relying on any particular interatomic potential, use a prototypical sixth order polynomial energy density developed in [85, 87] which ensures stress continuity over the whole configurational space. Specifically, we assume that

$$\varphi_0 \left(\frac{\tilde{\mathbf{C}}}{(\det \tilde{\mathbf{C}})^{1/2}} \right) = \beta \psi_1 \left(\frac{\tilde{\mathbf{C}}}{(\det \tilde{\mathbf{C}})^{1/2}} \right) + \psi_2 \left(\frac{\tilde{\mathbf{C}}}{(\det \tilde{\mathbf{C}})^{1/2}} \right), \quad (13)$$

where

$$\psi_1 = I_1^4 I_2 - \frac{41}{99} I_2^3 + \frac{7}{66} I_1 I_2 I_3 + \frac{1}{1056} I_3^2, \quad (14)$$

$$\psi_2 = \frac{4}{11} I_2^3 + I_1^3 I_3 - \frac{8}{11} I_1 I_2 I_3 + \frac{17}{528} I_3^2. \quad (15)$$

and we introduced the following hexagonal invariants of the metric tensor:

$$I_1 = \frac{1}{3}(\tilde{C}_{11} + \tilde{C}_{22} - \tilde{C}_{12}), \quad (16)$$

$$I_2 = \frac{1}{4}(\tilde{C}_{11} - \tilde{C}_{22})^2 + \frac{1}{12}(\tilde{C}_{11} + \tilde{C}_{22} - 4\tilde{C}_{12})^2, \quad (17)$$

$$I_3 = (\tilde{C}_{11} - \tilde{C}_{22})^2(\tilde{C}_{11} + \tilde{C}_{22} - 4\tilde{C}_{12}) - \frac{1}{9}(\tilde{C}_{11} + \tilde{C}_{22} - 4\tilde{C}_{12})^3. \quad (18)$$

The potential (13) contains a single parameter β that enables the enforcement of specific symmetries on the reference state. We select here $\beta = -1/4$ which ensures that global energy minimizers correspond to square lattices. The total elastic energy, allowing one to deal with both shear and volumetric deformations, is taken in the form

$$\varphi(\tilde{\mathbf{C}}) = \varphi_0 \left(\frac{\tilde{\mathbf{C}}}{(\det \tilde{\mathbf{C}})^{1/2}} \right) + \varphi_v(\det \tilde{\mathbf{C}}). \quad (19)$$

The volumetric part, which primarily influences dislocation core structures and controls the formation of voids, is assumed to be of a generic form

$$\varphi_v(s) = \mu(s - \log(s)) \quad (20)$$

which prevents infinite compression. To maintain in our numerical experiments the strain field near the surface $\det \mathbf{C} = 1$, we adopted a sufficiently high value of the bulk modulus, $\mu = 5$.

In Fig. 2(b) we illustrate the resulting energy landscape in the configurational space of metric tensors $\mathbf{C}$ with $\det \mathbf{C} = 1$. It highlights the fact that any such landscape would have a network of low energy valleys along the directions representing simple shears along crystallographic slip planes. It also reminds us that the deepest energy wells correspond to the equivalent replicas of the reference square lattice furnished by rank one deformation gradient tensors representing lattice invariant simple shears. Instead, pure shear deformations lead to extremely large elastic energy levels.

B. Finite element representation

The account for quantized lattice invariant shears makes the energy $\varphi(\mathbf{C})$ highly non convex. Therefore, the corresponding continuum (scale free) problem is highly degenerate. In particular, the energy minimization in such continuum setting can be expected to produce infinitely fine microstructures [104]. In fact, the lack of convexity of the potential is a property that the MTM shares with other similar Landau type continuum theories. The usual way of regularization in this case is the introduction of higher gradients (of the deformation, in our case) into the energy density and using the corresponding internal length scale as a cutoff parameter [105]. This would be an approach of a phase field theory which often leads to over-regularization and the loss of sufficiently fine aspects of the microstructure which is acceptable near critical points but may compromise the inherent nonlocality by treating it only in the long wave limit. Inside such a framework it is also a nontrivial task to represent adequately the global lattice symmetry. In view of these limitations, in the MTM approach we take a different path and bring an internal scale into the model through an explicit spatial discretization. Essentially, we reduce the space of admissible deformations to a finite dimensional set of compatible, piece-wise affine mappings.

More specifically, we assume that a crystal can be represented at the mesoscale by a collection of $N \times N$ discrete elements organized in a mesoscopic lattice which preserves the symmetry of the atomic lattice. The original lattice is effectively coarse-grained with an introduction of a uniform mesoscale grid. The deformation is assumed to be piecewise linear and the elastic response is attributed to discrete material elements whose linear size h is viewed as a physical parameter of the model. The numerical implementation of MTM is then based on solving an elastic finite element problem, with elements imitating mesoscopic aggregates of atomic cells. To trace the mechanical response of the crystal we follow the displacements of the network of discrete nodes labeled by integer valued coordinates $I = 1, \dots, N^2$.

We assume further that each of the elements is a deformable triangle and employ standard finite element discretization with linear triangular 3-node elements [106]. We begin by writing the displacement field inside each of these triangles in the form

$$\mathbf{u}(\mathbf{z}) = \sum_a \mathcal{N}^a(\mathbf{z}; h) \mathbf{u}^a, \quad (21)$$

where $\mathcal{N}^a(\mathbf{z}; \mathbf{h})$ are compactly supported linear shape functions, $\mathbf{u}^a$ are the amplitudes of the displacements of the nodes and summation is assumed over repeated indexes; the interpolation functions for each element are defined in terms of local dimensionless coordinate system. The mesoscopic deformation gradient is then

$$\mathbf{F}(\mathbf{z}; h) = \mathbb{I} + \mathbf{u}^a \otimes \nabla \mathcal{N}^a(\mathbf{z}; h). \quad (22)$$

In terms of macroscopic reference coordinates, the elastic energy inside each element can be computed using the simplest quadrature scheme

$$w(\mathbf{x}; h) = \frac{1}{2} \varphi(\mathbf{F}(\mathbf{x}; h)) J(\mathbf{x}, h), \quad (23)$$

where J is the determinant connecting local ($\mathbf{z}$) and global ($\mathbf{x}$) coordinate systems (**correct? CORRECT**).

We illustrate in Fig 4 our computational mesh which is a 2D lattice representing a scaled version of the original atomic lattice; a representative triangular finite element is, for instance, the element BAC adjacent to the node A. This element connects three nodes and therefore the index a associated with this element takes three values $a = 1, 2, 3$. We effectively assume in the MTM that the energy associated with the deformed element BAC depends on the three parameters: the deformed lengths of the bonds AB and AC and the deformed value of the angle BAC. To obtain these dependencies we use the Cauchy Born rule. Behind it is the assumption that the element deforms as if it was embedded into an infinite lattice undergoing a global affine (homogeneous) deformation. The latter is fully characterized by the same three parameters which we interpret as the components of the corresponding metric tensor C_{11} , C_{22} and C_{12} . Finding solution of an elastic problem implies local minimization of the

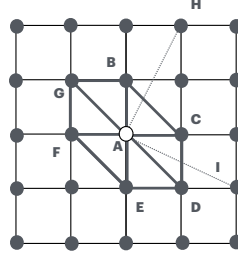


FIG. 4. Six triangular finite elements which are immediately adjacent to a generic node A.

energy

$$W = \int_{\Omega} w(\mathbf{x}; h) d\mathbf{x}, \quad (24)$$

which is prescribed on a triangulated domain Ω . The conditions of mechanical equilibrium, formulated in terms of the first Piola-Kirchhoff stress tensor

$$\mathbf{P} = 2\mathbf{F}\mathbf{m} \frac{\partial w}{\partial \mathbf{C}} \mathbf{m}^T, \quad (25)$$

take the form

$$\nabla \cdot \mathbf{P} = 0.$$

In the finite element representation these equations can be rewritten in the form

$$\frac{\partial W}{\partial \mathbf{u}^a} = \int_{\Omega} \mathbf{P}(\mathbf{x}; h) \nabla \mathcal{N}^a(\mathbf{x}; h) d\mathbf{x} = 0. \quad (26)$$

Referring again to our Fig 4, we observe that the elastic energy w dependence on the displacement of the node A has six contributions from the six adjacent triangular elements BAC, CAD, DAE, EAF, FAG and GAB. Therefore, in this setting, the node A interacts not only with all the nearest nodes B,C,E,F but also with some next to nearest neighbor nodes D and G. This interaction can be still interpreted as 'local', for instance, as it shown in Fig 4, the node A does not interact with mode distant nodes H and I. The implied 'multibody' interaction suggests that the energy of the node A depends not only on the distances to the corresponding nodes B,C,D, E, F, G but also on the three-point configurations represented by the angles BAC, CAD, AED, EAF and AFG. In other words, the force balance for the node A includes not only responses to the elongation of the bonds but also the associated three-body contributions.

We emphasize that such non-pairwise interaction is a result of coarse graining. In this sense the MTM approach is not in any contradiction with the fact that in the corresponding atomic model the interaction between the actual atoms is pairwise while maybe extending far beyond nearest neighbors. Essentially, in the MTM approach the energy of a single elastic element (triangle in our case) is computed as if the surrounding deformation field was affine which means that the local inhomogeneity of the deformation is underplayed.

Note also that the MTM, with parameter h equal to atomic scale and with nodes identified with actual atoms, can be seen as an atomistic scale model in which the actual inhomogeneity of local environment of a given atom is neglected even if the deformation of the element itself is properly accounted for. Such effective coarse graining allows can be used as a way to save computational time but at the expense of misrepresenting some microscopic aspects of the problem. Still, despite the inevitable shortcomings accompanying this type of approximations, the MTM approach remains advantageous as an accelerated computational scheme allowing one to bridge the gap between continuum and atomistic descriptions. The implied acceleration is due to the fact that one does not need to keep track of all neighboring atoms of a given element as the energy of the element can be computed in a straightforward manner based only on the information about the local value of the deformation gradient. The acceleration comes, of course, at a price: in the presence of an internal cutoff length scale some aspects of a genuinely atomistic description are necessarily lost.

The ensuing mathematical problem can be solved numerically using a quasi-Newton method accompanied by the Newton–Raphson (NR) "refinement" when the initial guess is too far from the solution for Newton–Raphson method to converge [107]. It has the advantage of being computationally efficient as well as being able to handle nonconvexity of the problem by only dealing with positive-definite Hessian matrices. The numerical method performs the minimization of the energy functional iteratively. In very general terms, it begins with an initial guess of $\mathbf{u}(\mathbf{x})$ and a given approximation of the *inverse* Hessian matrix H_0 as well as known optimal number m of BFGS corrections to be stored. An iteration first finds a minimization direction by performing a (one-dimensional) line search and finding the optimal step-size. Next, the solution is updated along with other quantities necessary to compute the inverse Hessian matrix of the next step. The algorithm differs from BFGS in the way it handles the inverse Hessian matrix, which is approximated/updated by m corrections at each iteration.

More specifically, in our numerical experiments to find $\mathbf{u}^a$, we first use the L-BFGS algorithm [108] which builds a positive definite linear approximation allowing one to make a quasi-Newton step lowering the total energy. The iterations continue till the increment in the advance of the total energy becomes sufficiently small. The obtained approximate solution is then used as an initial guess $\mathbf{w}^a$ to solve, using LU factorization [109, 110], the equations for the correction $d\mathbf{w}^a$ around an initial guess for the displacement field

$$\mathbf{u}^a = \mathbf{w}^a + d\mathbf{w}^a.$$

The corresponding system of linear equations can be written in the form

$$K_{ij}^{ab} dw_j^b + R_i^a = 0, \quad (27)$$

where

$$K_{ij}^{ab} = A_{ipjq}(\mathbf{F}) \frac{\partial \mathcal{N}^a}{\partial x_p} \frac{\partial \mathcal{N}^b}{\partial x_q}, \quad R_i^a = P_{ip}(\mathbf{F}) \frac{\partial \mathcal{N}^a}{\partial x_p}. \quad (28)$$

Note that here we used the Eulerian $i, j = 1, 2$ and the Lagrangian $K, L = 1, 2$ indexes and assumed summation on repeated indexes. We also defined

$$A_{iajb} = \frac{\partial^2 \varphi_D(\tilde{\mathbf{C}})}{\partial F_{ia} \partial F_{jb}}. \quad (29)$$

the tensor of tangential elastic moduli. We note that the knowledge of the Eulerian acoustic tensor

$$Q_{ij} = F_{ia} F_{mb} A_{iajb} n_i n_m, \quad (30)$$

where n are vectors of the unit circle allows one to formulate the criterion of an elastic instability of the corresponding continuum body in the form [111, 112]

$$\det(\mathbf{Q}) = 0. \quad (31)$$

In our numerical experiments we started with a pristine (defect free) square crystal sample represented by $N \times N$ finite elements with $N = 400$. We then loaded this system quasi-statically along one of the the principal slip direction by applying an affine displacement field representing hard device loading. Specifically, we prescribed on the boundary of our square sample $\partial\Omega$ the displacement field

$$\mathbf{u}(\alpha, \mathbf{x}) = (\bar{\mathbf{F}}(\alpha) - \mathbb{1})\mathbf{x} \quad (32)$$

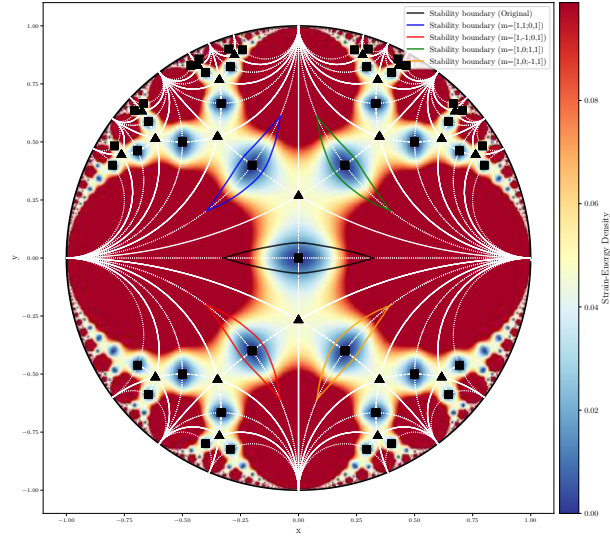


FIG. 5. The stability envelope superimposed on the strain-energy density landscape in metric space visualized via Poincaré (stereographic) projection. The color map represents the strain-energy density field. The stability boundaries, derived from acoustic tensor analysis where $\det(\mathbf{Q}) = 0$, enclose the region where homogeneous solutions are mechanically stable. Five boundaries are shown: the original stability envelope and four copies generated by applying the coordinate transformations $\mathbf{m}^T \cdot \mathbf{C} \cdot \mathbf{m}$ with $\mathbf{m} \in \left\{ \begin{bmatrix} 1 & \pm 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ \pm 1 & 1 \end{bmatrix} \right\}$, representing equivalent energy wells in the metric space.

which was applied to all boundary nodes. The special applied deformation gradient

$$\bar{\mathbf{F}}(\alpha) = \begin{bmatrix} 1 & \alpha \\ 0 & 1 \end{bmatrix} \quad (33)$$

was chosen to represent simple shear along an “easy” glide direction with shear amplitude α serving as the loading parameter. By changing this parameter in increments of order 10^{-6} , we incrementally advanced the loading, while waiting for the system stabilization after each loading increment. Specifically, after each incremental advance of the parameter α , the displacement field was updated by our energy minimization algorithm which used the solution from the previous step as the initial condition. As we have seen, the energy minimization required a succession of iterations which is natural to interpret as a ‘fast-time’ overdamped dynamic process taking place at a fixed value of the loading parameter. The latter is then interpreted as evolving in a ‘slow-time’ scale. We effectively assumed full time scale separation allowing the ‘fast-time’ internal iterations to continue at the given value of the loading parameter till the value of the energy increment reached below a sufficiently small threshold. After the termination of the energy relaxation stage, the loading parameter was advanced again.

C. General comments

Before we turn to the results of our numerical experiments, some general comments about the MTM approach may be suitable as the physical implications of these experiments depend on the chosen modeling framework.

Recall first that in molecular dynamics approach (or rather molecular statics (MS) in our quasi-static loading case) there is also a characteristic internal scale: the interatomic distance a which is a microscale parameter. In the mesoscopic approach of the MTM the coarse-graining scale h , defining the cutoff beyond which the deformation is considered homogeneous (affine), is necessarily larger than a since the goal is to accelerate the computations. Their relative magnitudes should be, of course, discussed in terms of dimensionless quantities. Thus, if the size of the macroscopic domain is L , its division into finite elements introduces a dimensionless internal scale $h = L/N$ where we recall that N^2 is the number of the nodes in the mesoscopic finite-element grid. The analogous microscale parameter is $\epsilon = a/L$ and while we are ultimately implying the limit $\epsilon \rightarrow 0$, $\delta \rightarrow 0$, we must assume that $\delta \gg \epsilon$. To locate in this perspective the classical continuum approach, we recall that h can be interpreted as an effective size of a ‘continuum particle’. The, to obtain the continuum limit we first implicitly perform the limit $\epsilon \rightarrow 0$ and recover in this way local constitutive response by using, for instance, the Cauchy-Born rule. Then, to obtain a scale free theory we perform the second limit $\delta \rightarrow 0$. Essentially we consider a double asymptotics $\epsilon \rightarrow 0$, $\delta \rightarrow 0$, with $\delta \gg \epsilon$ which implies that $a \ll h \ll L$ and the only remaining length scale is L , see Fig. 6. Instead, in the MTM approach we effectively consider only

a single limit $\epsilon \rightarrow 0$ which allows us still to use the Cauchy-Born rule. We then avoid the second limit and maintain a small but finite value of the parameter δ which then necessarily satisfies the constraint $\delta \gg \epsilon$. In this situation we have $a \ll h \sim L$ and both h and L survive in the limit $a/L \rightarrow 0$, see Fig. 6. In this sense the MTM is essentially a continuum theory with an internal length scale, like, say, phase field model and other similar ‘quasi-continuum’ approaches.

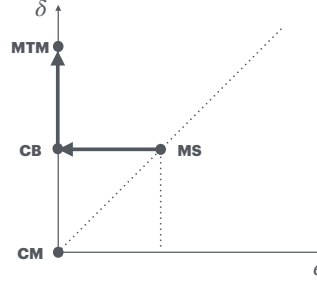


FIG. 6. Schematic representation of the asymptotic ($\epsilon \rightarrow 0$, $\delta \rightarrow 0$) inter-relations between the molecular statics (MS), the continuum mechanics (CM) and the mesoscopic tensorial model (MTM). As the (normalized) atomic scale ϵ tends to zero, the elasticity can be approximated by the energy density obtained through the application of the Cauchy-Born rule (CB). If the mesoscopic scale δ is eliminated, we obtain CM and if it is kept finite and larger than the atomic scale, we obtain MTM.

As on all such theories, making the dimensional parameter h smaller is equivalent to making another dimensional parameter L larger. In this sense the assumption that δ is finite is equivalent to incorporating finite size effects. Thus, even though changes in δ will scale the size of the core regions of dislocations, they will preserve the nature of their long range interactions, at least far away from the boundaries. The use of Cauchy-Born coarse grained description, will also lead to some loss of microscopic details in the description of dislocation cores, however, the ensuing ‘blurred’ dislocations will still nucleate and self-lock adequately even if perhaps at different values of controlling parameters than in the corresponding MS simulations. Note also that at least some of the characteristic parameters including the core energy, the Peierls stress, the elasticity of a dislocation line, etc., may be matched by using δ dependent interatomic potential which will then affect the barrier structure in the corresponding Cauchy-Born energy density. The situation here is the same as in the case of phase field description of phase boundaries where the need to reproduce the realistic value of the surface energy requires the microscopic potential to be appropriately re-scaled.

Given the nature of the approximation provided by the MTM approach, it can be compared to other physical models with partial spatial resolution. An important example is the model of turbulence known as large eddy simulation (LES), e.g. [113, 114]. This computational approach deals with a similar problem of choosing an ultraviolet cutoff while describing the processes at the smaller scales by effective closure schemes. The principal idea behind LES is the same as in the MTM approach: to reduce the computational cost by ignoring the smallest length scales which turn out to be the most computationally expensive to resolve. In both LES and MTM, the omitted small scale details are mostly sacrificed in the interests of addressing the scale invariant aspects of the phenomena. As in the MTM, the sub-grid scales in the LES are not irrelevant, and their effect is modeled by sub-grid scale models which are not rigorously derived from the micro-scale but, instead, combine empirical information with some fundamental physical constraints. The choice of the grid size in LES is rooted in the idea that LES accurately captures the energy-containing large eddies while modeling the smaller ones only effectively to maintain a balance between accuracy and computational feasibility. Similarly, the choice of the cutoff scale h in the MTM, where the sub-grid description reduces to Cauchy-Born type representation of the energy density, is guided by the necessity to capture within available computational resources the collective effects involving a large number of dislocations leading to intermittent avalanche behavior and attendant spatial complexity. In this perspective, the actual value of the parameter h in the MTM approach is irrelevant as long as the *quantitative* comparison with MS is not an issue and the computational domain is large enough so that the finite size effects due to interaction of enlarged dislocation cores with external boundary can be still neglected. In such settings, by changing h we end up capturing the same physical interactions even if they are viewed with different degree of resolution.

Therefore, we adopt a pragmatic approach to the choice of the mesoscopic length scale h and make it problem-specific. The idea is to ensure that the Cauchy-Born energy density, computed for a *finite* lattice fragment with scale h , can be considered as approximately periodic within the interesting range of strains. In other words, we would like to capture correctly the targeted energy wells while probably sacrificing the $GL(2, \mathbb{Z})$ periodicity beyond the actually reachable range. Since the exact global periodicity is recovered fully in the Cauchy-Born approach only in the limit of infinite sample size, our strategy suggests that the cutoff scale h should be increased when the goal is to capture the phenomenon of progressively larger slips along the same slip system. **(which value of h are we actually choosing? we take $h = 1$)**

Finally, it is important to stress that despite its unusual, for plasticity theory, ‘nonlinear elasticity’ appearance, the MTM approach is conceptually very close to the classical continuum plasticity (CP). For instance, both theories effectively account for the complexity of energy landscape by introducing low-energy valleys describing plastic ‘mechanisms’. However, if in CP, additional phenomenological relations are needed to handle the coupling between different ‘mechanisms’ [115–119] in the MTM

approach, such coupling is automatic as the model has access to the height of the energy barriers separating different low-energy valleys. It also accounts for the barriers inside such valleys, moreover the finite element representation, creates an even broader set of metastable states. A mechanical system driven elastically in the ensuing 'bumpy' energy landscape experiences a rich repertoire of snap-through instabilities which are ultimately behind the dissipative nature of the MTM. Indeed, as we have already seen, such instabilities are described implicitly by an overdamped model with relaxation time much shorter than the characteristic time of the loading. Therefore the actual dissipative process is hidden inside the local energy minimization protocol which hides rate independent plasticity behind the nonlinear elasticity appearance.

III. NUMERICAL EXPERIMENTS

A. Periodic Boundary Conditions with Macroscopic Deformation

To simulate an effectively infinite crystal under macroscopic deformation while maintaining computational tractability, we implement periodic boundary conditions (PBCs) through domain replication. The computational domain Ω_0 with dimensions $L_x \times L_y$ is replicated in all spatial directions to create a periodic array of domains.

1. Translation Vectors and Domain Geometry

We define a set of translation vectors $\mathbf{t}_n$ that generate the periodic lattice of domains:

$$\mathbf{t}_n = \begin{pmatrix} n_x L_x \\ n_y L_y \end{pmatrix}, \quad \mathbf{n} = (n_x, n_y) \in \{-1, 0, 1\}^2,$$

where $\mathbf{n}$ represents the integer lattice coordinates. For the 9-domain implementation, we consider all translation vectors within the immediate neighborhood of the central domain.

Let $\mathbf{F}(t)$ denote the macroscopic deformation gradient tensor at time t . The position of any replicated domain relative to the central domain Ω_0 is given by:

$$\mathbf{R}_n = \mathbf{F} \cdot \mathbf{t}_n,$$

where $\mathbf{R}_n$ represents the displacement vector of the origin of the domain corresponding to translation $\mathbf{t}_n$.

2. Particle Position Mapping

For a particle located at position $\mathbf{x}_i$ in the central domain Ω_0 , its periodic image corresponding to translation vector $\mathbf{t}_n$ is positioned at:

$$\mathbf{x}_i^{(n)} = \mathbf{x}_i + \mathbf{R}_n = \mathbf{x}_i + \mathbf{F} \cdot \mathbf{t}_n.$$

This ensures that the macroscopic deformation gradient is consistently applied across all periodic images, maintaining the correct strain field representation.

3. Delaunay Triangulation and Triangle Selection

After establishing the positions of all particles across the nine domains (original domain Ω_0 and its eight periodic images), we perform a Delaunay triangulation on the complete set of particles:

$$\mathcal{P} = \{\mathbf{x}_i : i \in \mathcal{I}_0\} \cup \{\mathbf{x}_i^{(n)} : i \in \mathcal{I}_0, n \neq \mathbf{0}\},$$

where $\mathcal{I}_0$ denotes the set of particle indices in the original domain and $\mathbf{0} = (0, 0)$ corresponds to the null translation vector. This means we deal with the all particles in the original central domain and all periodic images of those particles in the 8 surrounding domains.

From the resulting triangulation $\mathcal{T}$, we select only the unique triangles that have at least one vertex belonging to the original domain Ω_0 . The initial selection of all triangles with at least one node connected to the original domain may introduce duplicated triangles due to the periodic replication. Therefore, we apply an additional filtering step to remove these duplicates and retain only unique triangles. This selection and filtering procedure ensures that:

1. Each triangle in the computational mesh is counted exactly once, eliminating redundancy from the periodic duplication
2. All triangles connected to the original domain are included, maintaining proper connectivity across periodic boundaries
3. The triangulation naturally incorporates the macroscopic deformation through the deformed positions of the periodic images

The selected set of triangles $\mathcal{T}_{\text{unique}}$ forms the computational mesh for the mesoscopic tensorial model calculations, preserving both the periodic boundary conditions and the consistency with the applied macroscopic deformation gradient $\mathbf{F}$.

In view of the pristine nature of our samples, the initial phase of the deformation process, starting at point A in Fig. 7, is necessarily purely elastic. During this stage, the metric tensors of all elements remain strictly inside the elastic domain (Pitteri neighborhood). While the response of the crystal inside such a domain could, in principle, remain homogeneous, the affine deformation becomes unstable way before its boundary is reached. The implied purely elastic instability is preceded by the elastic softening and the actual instability takes place near the maximal load. The corresponding critical value of the loading parameter $\alpha = \alpha_c \approx 0.138$ can be well approximated if we assume that the instability is associated with the loss of positive definiteness of the acoustic tensor (30). As we have already mentioned, the corresponding Legendre-Hadamard conditions [120–122], indicating that the equilibrium equations of continuum elasticity stop being strongly elliptic at the corresponding affine configuration, are represented by (33). However, in view of the finite element nature of our actual problem the solution of (33) provides only an approximation of the critical value of the loading parameter with the corresponding correction being of order $(h/L)^2 = N^{-2}$. Note also that in view of an pristine nature of the crystal, at the critical value of the loading parameter (point B in Fig. 7) all finite elements can be expected to become unstable simultaneously. This would lead to a nonphysical highly unstable behavior and therefore, to observe a generic response we had to regularize the problem by adding at the onset of instability a small amount of quenched disorder which was removed right after the instability took place.

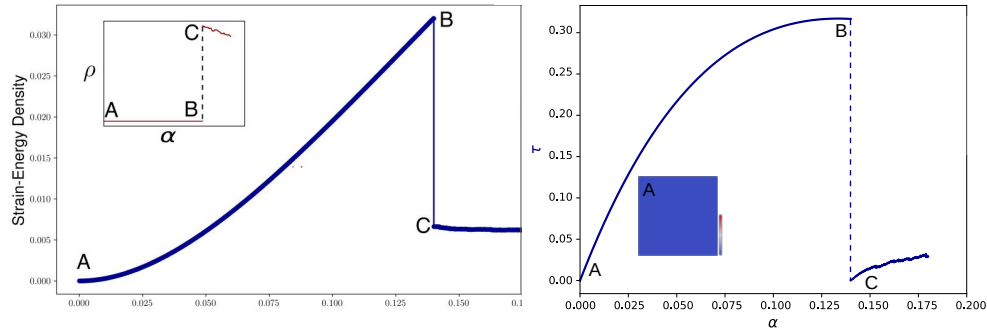


FIG. 7. (**change the order of a and b. to be done later**) Priming of the pristine sample and first stage yielding. (a) Strain-energy density evolution during proportional loading. The labeled configurations are A (initial), B (first yielding point), and C (post yield). The sharp drop of energy from B to C represents the transition from pure crystal to an 'amorphous crystal', see the text for more details. The subsequent microplasticity regime takes place at almost constant energy. Colors in the inset indicate the level of strain-energy density. (**correct? YES**). (b) The shear component of the Cauchy stress tensor field τ evolution along the same loading path. The instability takes place when the crystal reaches its theoretical strength threshold. The stress drops almost to zero opening the way to a hardening regime of 'microplasticity'. The inset illustrates the evolution of the corresponding dislocation density ρ . Here $N = 400$.

The overall macroscopic mechanical response before and during the instability is illustrated in Fig. 7(a,b). As we have already said, the loading process begins in point A where at zero applied strain (**explain which stress component is plotted here? WE always plot The shear component of the Cauchy stress tensor field**) the pristine crystal is undeformed and has zero elastic energy. The configuration continues to be homogeneous as the loading increases and we observe the nonlinear softening elastic behavior dictated by the chosen nonlinear elastic potential. Near the load maximum of the stress-strain curve (point B) we introduce a small quenched disorder. The resulting pre-instability highly stressed configuration is shown in Fig. 8(a) where we show the energy distribution inside individual finite elements. Note that this state is almost homogeneous, in particular, the imposed small stress modulation does not result in the formation of dislocations.

Our next Fig. 8(b) shows the post-instability configuration (point C). One can see that the breakdown of an almost affine elastic state takes the form of a major system-size rearrangement transforming a perfect crystal into a dislocation-rich, apparently 'amorphous crystal'. Note that in the course of this catastrophic system-spanning event the crystal abruptly loses almost all its accumulated elastic energy with stress dropping almost to zero, see Fig.7 (a,b). The inset in Fig.7(a) shows the jump of the dislocation density ρ from zero in both configurations A and B to a finite value in the state C .

The deformation pattern emerging after the stress drop (point C), reveals *collective* nucleation of a large (**how many?**) number of dislocations ≈ 2000 . The latter were identified using the classical Delaunay triangulation method. It allowed us to detect the coordination number anomalies and $5/7$ coordination defect pairs were interpreted as dislocations. One can say that an artificially

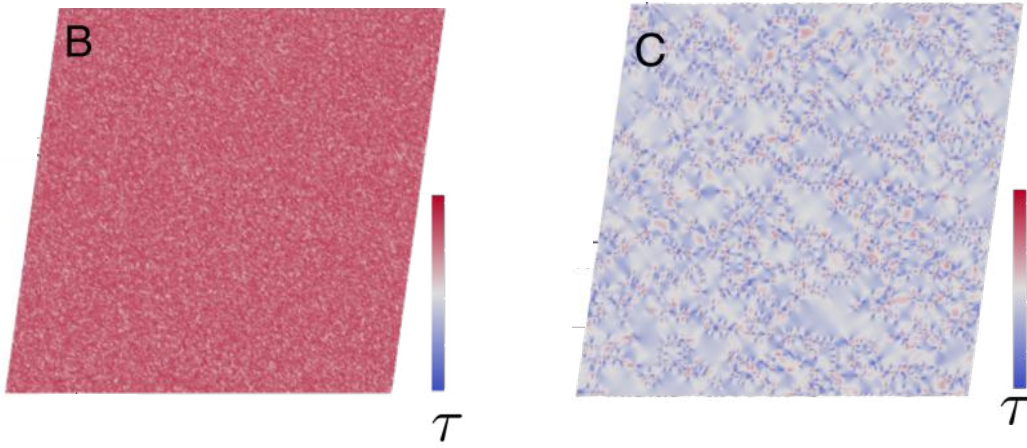


FIG. 8. Emergence of a quasi-glassy state. (a) Highly stressed perfect crystal close to the first yield point (B) with small initial quenched disorder introduced to break the unphysical degeneracy of the pre-yielding state. (b) Post- (first)yielding configuration (C) after massive nucleation of dislocations which self-organize into an intricate multi-scale pattern. Colors indicate the level of energy density. (correct?)

created defect-free (cold) sample was converted into a generic (hot) sample carrying a realistic initial distribution of defects, see Fig. 8(b). We call such ‘preparation’ or ‘priming’ realistic because the defects are not placed by hand but instead emerge in the course of self organization. The corresponding highly cooperative event unfolded dynamically in a fast computational time involved in local energy minimization. In other words, one can say that the achieved configuration is the first stabilized state reached by an overdamped relaxational dynamics involving nucleation and annihilation of dislocations as well as the formation of dislocational entanglements. It is then natural to expect that the attained state is only marginally stable which is an observation which we discuss in more detail in what follows.

Note next that while the observed catastrophic mechanical response is still associated with *plastic* deformation, its discontinuous nature is highly reminiscent of brittle fracture. The phenomenon of plastic brittleness in crystal plasticity is realistic, moreover it has been qualified as a characteristic feature of sub-micron crystal samples with high initial purity (absence of solutes, precipitates, and dislocations) [123–133]. However, as we discussed in the Introduction, such discontinuous yield is also reminiscent of the mechanical response of well-annealed glasses. In particular, we have seen that among other amorphous systems which exhibit similar quasi-brittle behavior are there are jammed configurations of granular media which are only marginally stable. In this perspective, the dynamic process, transforming the pre-avalanche configuration *B* into the post-avalanche configuration *C*, can be viewed as converting an unstable ‘fluidized’ system into a marginally stable one. In view of its highly disordered appearance, which of course does not indicate the absence of correlations, and in the absence of other established choices, we refer to this state as the ‘amorphous crystal’.

After such *preparation* stage, we continue to apply the same athermal quasi-static loading protocol to the emerging ‘glassy’ system. Specifically, we subject our highly dislocated crystal to progressively larger simple shears. The ensuing stress-strain response is illustrated in Fig. 9. Its salient feature is the presence of irregularly placed elastic branches interrupted by abrupt stress drops. The latter represent intermittent plastic avalanches which involve partial or complete unlocking of dislocational structures; the associated dynamic restructuring of the deformation pattern may be either local or global.

In Fig. 9 we can identify two rather different regimes. The first regime shows plastic hardening behavior and is represented by the configurations *D*, *E* and *F*. In this regime the dislocation density remains almost constant. Then there is a localized major multi-stage stress drop represented by the transition $F \rightarrow G \rightarrow H$. During the associated system spanning restructuring event the dislocation density abruptly jumps. Finally there is a quasi-stationary regime of an ‘easy glide type’ represented by a stress plateau and exemplified by the state *I*. According Fig. 9, in the ‘hardening’ regime the dislocation density remains almost constant. In this regime the dislocation density progressively increases.

The observed yielding behavior is highly reminiscent of the plastic response of well annealed ‘brittle’ glasses. To understand it better we now turn to the analysis of the corresponding spatial dislocational micro-configurations.

In Fig. 10 we show the energy distributions in the states *D* and *E*; note that the highest energy levels correspond to the location of dislocation cores. One can see that during the transition $D \rightarrow E$ no grain boundaries or shear bands, appear to be forming. Only preexisting mobile dislocations are moving between the locking sites formed by immobilized dislocations. One can say that dislocations, which remain mobile after the first massive dislocation nucleation event, are confined inside the effective cages where they can pile up but from where they can only rarely escape by breaking the existing locks.

Note next that in this regime both major slip systems are activated and small dislocation avalanches, corresponding to weakly coordinated dislocation motions along their gliding planes, are abundant. The characteristic entanglements involve

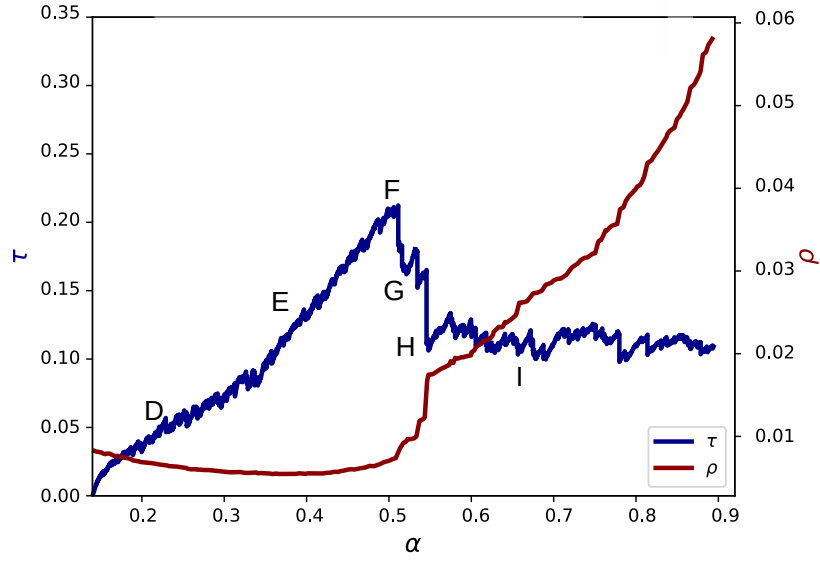


FIG. 9. Post- (first) yielding deformation history: a typical stress vs. strain curve and the corresponding dislocation density in a system with $N = 400$. The hardening stage from D to F shows the pre-(second) yield 'microplastic' deformation. The transition from F to H correspond to a cluster of system spanning avalanches. The post-(second) yield stationary plasticity regime is represented by the state I on a stress plateau.

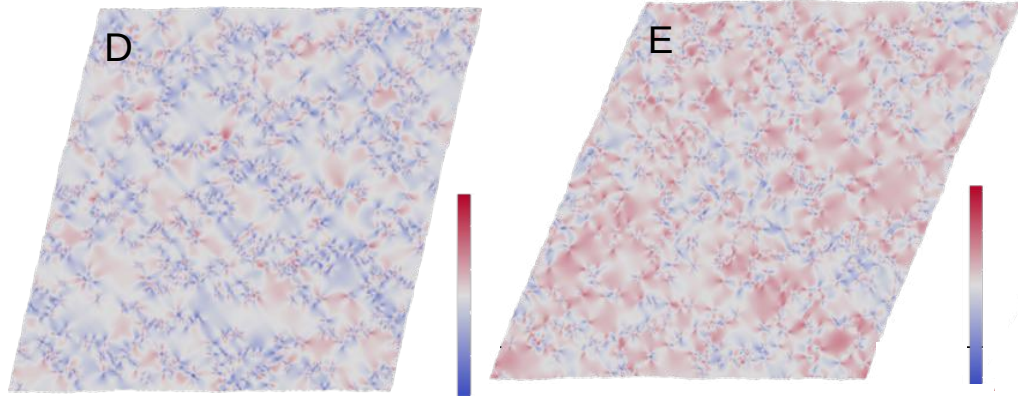


FIG. 10. Post-(first)yielding which is also a pre-(second) yield deformation history: hardening regime representing 'microplasticity' with almost constant dislocation density and with major dislocation self-locks (entanglements) between the states D and E remaining intact. Colors indicate the magnitude of shear stress component τ_{xy} of the Cauchy stress tensor.

dislocations on two different slip planes transiently blocking each other. The processes of nucleation and annihilation of dislocations motion appear to be largely suppressed. In Fig. 8 we see that even though beyond the state C the system exhibits stress-hardening behavior, the energy remains almost unchanged. This means that the work of the loading device is almost all converted into heat as the dislocation number remains basically fixed.

Overall, such pseudo-elastic regime preceding macroscopic yield can be interpreted as micro-plasticity. It is represented by local plastic events that individually do not lead to statistically meaningful changes in dislocation structure and are usually not resolved in conventional bulk deformation experiments. However, this small-scale intermittent micro-plastic activity which usually becomes apparent only in samples of reduced size, is still characteristic of pre-yield behavior for both crystal and glassy systems [4, 134–138].

We now turn to the macroscopic yield which we associate with the second major restructuring event starting at point F . The whole system-size avalanche takes the form of a two-stage transition $F \rightarrow G \rightarrow H$. The corresponding snapshots of the spatial dislocational micro-configurations are shown in Fig. 11. Here we already see the evidence of a collective activity of dislocations which leads to the formation of global structures. This trend is mostly pronounced during the abrupt stress drop associated with the transition $G \rightarrow H$ where we observe massive nucleation of dislocations bringing about a system-spanning shear band.

Our Fig. 12, presenting the snapshots of the strain-energy density field in the states F and G , illustrates in more detail the

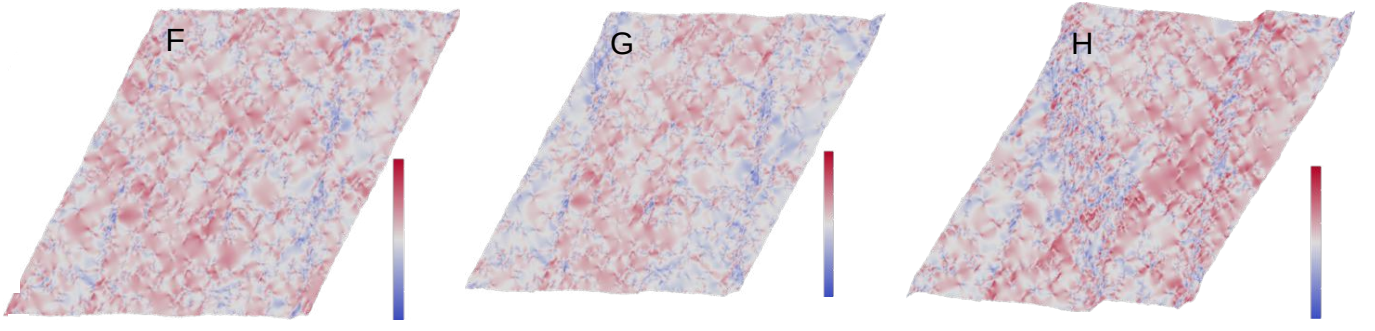


FIG. 11. Second stage yielding represented by the system size restructuring of a dislocation pattern. Major shear band forms between the states G and H . Colors indicate the level of energy density. (correct?)

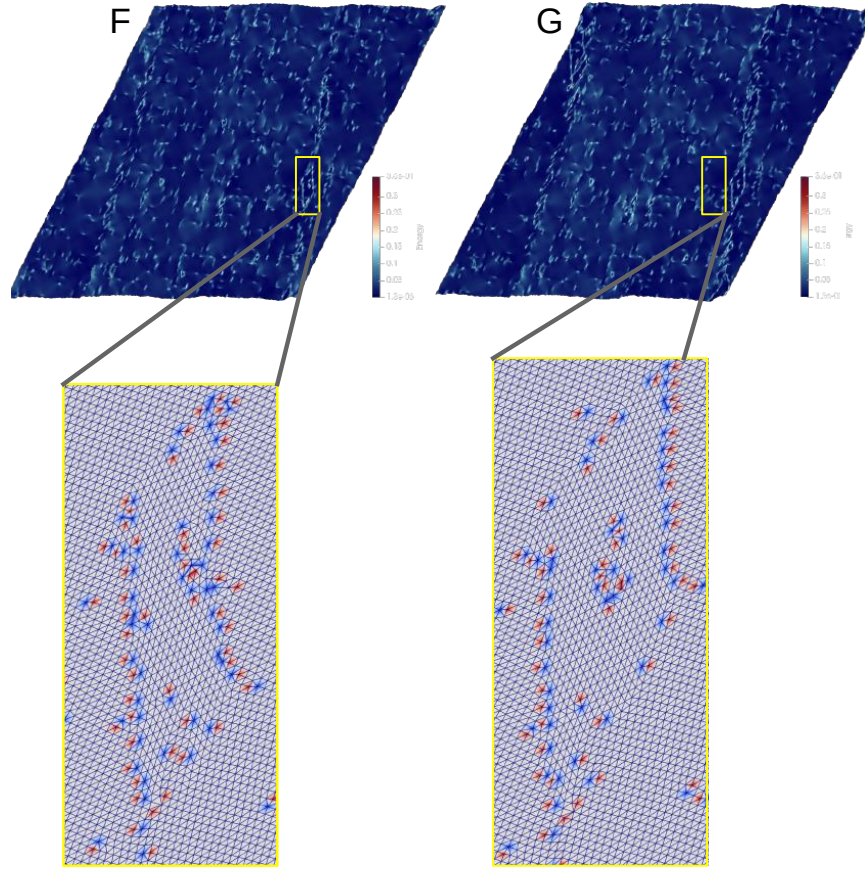


FIG. 12. **(Correct the location of the yellow box in (b) to done later)** Evolution of the shear component τ_{xy} of the Cauchy stress tensor field and, the associated changes in the dislocation structures during the transition between the states F and G . Yellow rectangles show the microstructural evolution of the same set of nodes. We use here Delaunay triangulation, with 5/7 coordination defect pairs indicating dislocation positions. The formation of grain boundaries at the (second) yield point is followed by the their propagation in the process of the elongation of a crystal grain with fixed misorientation. Different grains appear to be self-organizing into a system spanning shear band.

first stage of this critical transition. Here again the regions with higher energy are localized around dislocation cores. To reveal the underlying microstructural mechanism, responsible for the macroscopic stress drop, we visualized Delaunay triangulation in selected regions which are highlighted by yellow rectangles; as we have already mentioned, this allows one to track through the coordination number anomalies the evolution of dislocation configuration.

According to the collected information, the yield point (state F) marks the initial formation of grain boundaries which emerge as a result of self-organization among a large number of dislocations. Such self organization becomes possible due to the

presence of both, long range elastic interactions and short range interactions induced by dislocation cores. The implied incipient development of the grain texture marks the initial shift from 'local' to 'global' dislocational arrangements. During the subsequent gradual unfolding of the system spanning dislocational avalanche, exemplified by the state G , the microstructural evolution is dominated by the propagation and growth of the previously created grain boundaries. Finally, an even more radical rearrangement of the dislocational structure, bringing about the state H , marks the coalescence of different grains and points towards the formation of at least one extended shear band. One can view the emerging bands as representing collections of

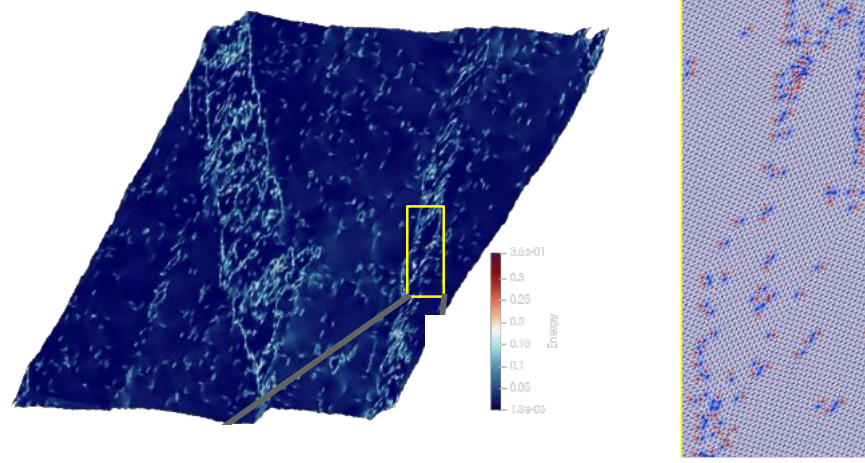


FIG. 13. The post-(second) yield configuration corresponding to state H is presented through the corresponding stress field. Multi-grain texture can be seen behind the system spanning shear bands.

differently oriented grains. The corresponding mis-orientations are controlled by the overall compatibility of the deformation field. In particular, the ubiquitous presence in such patterns of low energy $\Sigma 5$ grain boundaries is discussed in detail in [16], see also [139–141].

The emerging low energy grain boundaries are still dislocation-rich which explains the necessity to nucleate a large amount of dislocations during the transition from state G to state H . Such dislocational arrangements are highly optimized which suggests that at this stage of the deformation history the self-organization becomes global.

We show in Fig. 13 a representative snapshot of the micro-configuration of the crystal in state H . It reveals that the avalanche $G \rightarrow H$ involves both, a growth of an existing grain and its merger with another grain. This consolidates the structure of the shear band which may be also viewed as an element of a polycrystalline texture. The corresponding catastrophic stress drop may be explained by the fact that, in view of their highly optimized nature, stresses required to move grain boundaries are smaller than those sufficient to move individual dislocations.

We reiterate that we interpret the progression $F \rightarrow G \rightarrow H$ as the transition from quasi-elastic, pre-yield behavior of an 'amorphous crystal' to the actual plastic yield. We stress again that it is highly reminiscent of brittle yielding in well-annealed glasses. The analogy is, of course, incomplete since, for instance, in the case of glasses the dislocational interpretation of a structure of the shear band would be inappropriate.

The post-yield fully plastic yielding, represented by the states H and I , can be interpreted as quasi-stationary non-equilibrium steady state due to relative stabilization of the average stress level. A representative snapshot of the micro-configuration of the crystal in state I is shown in Fig. 14. The intermittent avalanches, characterizing this regime, represent the subsequent broadening and development of the polycrystalline grain texture. While bigger grains occasionally merge to form larger rotated patches, small grains continue to nucleate which requires an increase of the number of dislocations. In this way the self-organization of dislocations within the existing shear bands continues while the latter keep widening. Overall, despite the increase in the number of dislocation, we observe continuous maturation of what appears to be a scale free hierarchical structure while the average macroscopic stress remains constant.

As the system transitions from isolated dislocation motion to collective dislocational behavior, it gains access to a much broader repertoire of large-scale relaxation mechanisms, in particular, the ones that can release greater amounts of energy in a single avalanche event. The transition from pre-yield to post-yield behavior can be then interpreted as a result of internal optimization allowing the system to cope with the necessity to dissipate larger amount of energy. To illustrate the fundamental difference in the energy dissipation mechanisms before (states D , E and F) and after (states H and I) the yielding transition, we show in Fig. 15 the typical spatial distributions of slipped regions during a single avalanche event for both pre- and post yielding regimes.

Using the language of amorphous plasticity, one can say that in the pre-yield regime, see Fig. 15(a), plastification occurs primarily in isolated, linear arrangements of transformed elements showing minimal branching. These structures correspond

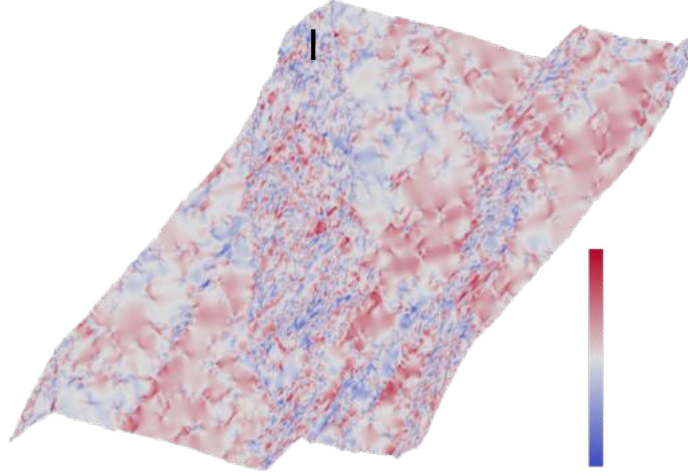


FIG. 14. Mature shear bands in the regime of stationary post-(second) yield plastic flow represented by the state I . Colors indicate the levels of energy density (correct?).

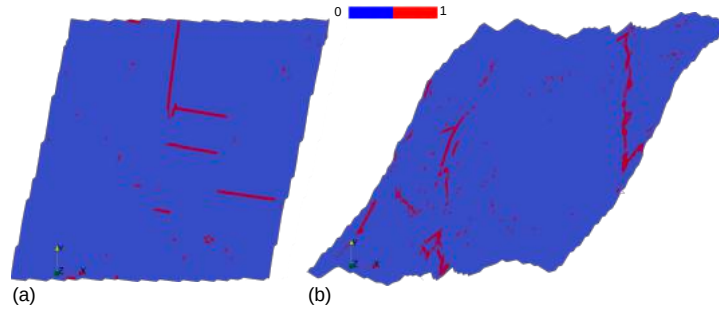


FIG. 15. The configuration of plastically activated regions (shown in red) associated with typical stress drops: (a) pre-(second) yield avalanche ; (b) post-(second) yield avalanche.

to individual dislocations advancing along one of the two crystallographic planes. In this case the energy dissipation is highly localized while the slip pattern is sparse, reflecting a weak nature of dislocation interaction.

The configuration of a typical post-yield avalanche reveals a different pattern of zones of plastic activity, see Fig. 15(b). Here, we observe extended plastified regions showing complex branching. These structures emerge as a result of the correlated nucleation and migration of grain boundaries producing a texture of almost unloaded purely crystalline patches which differ only by their orientation. These observations confirm that in the post-yield regime the necessity of efficient energy dissipation indeed enforces a much more elaborate spatial structuring. As we have already mentioned, the implied complexity is achieved as a result of the transition from individual dislocation motion to their collective self-organization.

IV. STATISTIC OF AVALANCHES

We finally turn to the analysis of statistics of plastic avalanches s in our numerical experiments. Fluctuational activity has been lately realized as representing an important part of the mechanical response of crystals undergoing plastic deformation [1, 4, 11, 77, 142–144]. Our analysis will concern pristine crystals, but only after they have already undergone the initial elastic instability transforming them into ‘amorphous crystals’. As we have seen, after such a ‘preparation’ an initially stable system is effectively quenched in apparently marginally stable state. In other words, we focus on what we call the ‘second’ plastic yield. The goal is to investigate the nature of plastic fluctuations by computing the probability distribution of the energy drops associated with individual avalanches in pre-(second)yield regime ($\alpha < \alpha_c$) and post-(second)yield regime ($\alpha > \alpha_c$).

Before we report our results, it is appropriate to mention again that in amorphous solids the statistics of similar energy drops has been found to be represented by a power law with a cut-off that depends on system size [43, 145–147]. We also recall that in the context of amorphous plasticity such behavior is usually described in terms of simple elasto-plastic models, in which linear elastic finite elements can yield above some critical stress and then trigger the yield of nearby elements as the stress is

transmitted through the system by an elastic propagator [33, 148–150]. We reiterate that the MTM approach to crystal plasticity operates within very similar finite element setting. However, there are also some important differences. Thus, phenomenological yield thresholds are replaced by elastic instabilities in a given energy landscape and fixed elastic propagators are replaced by the solution of a nonlinear elasticity problem. It is also important the MTM approach to deformation is geometrically exact which allows one to handle adequately finite rotations and elastic compatibility.

Despite these differences, our analysis shows that in both pre- and post-yield phases the statistics of intermittent energy-dissipating events accompanying plastic deformation closely resembles the behavior of similar characteristics in well-annealed glasses. Specifically, we found that the probability density of energy drops in our numerical experiments can be well approximated by the power-law distribution with an exponential cutoff

$$P(x) \sim x^{-\epsilon} e^{-\lambda x}, \quad (34)$$

is exactly the result obtained in the studies of plasticity in amorphous solids [4, 28, 62, 76, 144, 151].

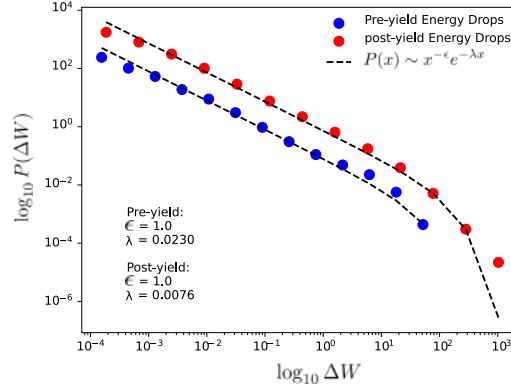


FIG. 16. Probability distribution of energy drops (ΔW) during avalanches in pre-(second) yield regime (blue points) and post-(second) yield regime (red points). Both distributions are well approximated by a power-law with exponential cutoff. The power law exponents ϵ in both regimes are identical. Instead the cutoff parameters λ are drastically different.

Our own results are summarized in Fig. 16 where we compare the statistics of both pre- and post-(second)yield avalanches. One can see that in both cases the power law range is very stable extending over up to six decades. Remarkably, despite all the dramatic differences in the macroscopic behavior, both pre-yield and post-yield energy distributions share the same power-law exponent: $\epsilon = 1.0$. This is an indication that the scaling behavior remains basically unchanged across the yielding transition despite the fact that we observe hardening in the pre-yield regime where the dislocation number remains basically unchanged and stationary plastic flow in the post-yield regime, where hardening disappears but the dislocation number continuously increases. The invariance of ϵ across the yielding transition suggests that while the spatial organization of plastic events changes drastically as a result of yielding transition from local and non-cooperative to global and collective, the statistical distribution of events in the critical regime remains insensitive to these microscopic details revealing a bigger picture of the global organization of the underlying energy landscape [20, 152]. The universality of the obtained result is corroborated by similar findings using a completely different choice for the $GL(2, \mathbb{Z})$ energy density function [14].

The value of the power-law exponent $\epsilon = 1.0$ has been encountered in the studies of crystal plasticity even if in somewhat different contexts. Thus, it was found to be a signature of an archetypically “wild” crystal plasticity in the sense of [77].

An almost the same value of the power-law exponent $\epsilon = 1.0$ was obtained in discrete dislocation dynamics (DDD) studies. In their numerical experiments conducted on disorder-free 2D samples containing a fixed number of preexisting dislocations the authors associated exactly this value of the power-law exponent with extended range of steady plastic flow [62, 153]. The exponent $\epsilon = 1.0$ was also recorded in numerical experiments involving pristine crystals conducted using the scalar version of the MTM [17]. Overall, in the literature on crystal plasticity the exponent $\epsilon = 1.0$ has been previously linked with either dislocation jamming or self-induced glassiness [28, 154–156].

Some theoretical understanding of the relation between the emergence of the power law exponent value $\epsilon = 1.0$ and the structure of the underlying energy landscapes has been obtained in the case of spin glasses. Thus, it was shown that the hierarchical (ultrametric) organization of energy wells in the phase space results in an intermittent, scale-free response to quasistatic deformation [157–160]. Moreover, it was shown that in the mean-field limit the associated distribution of the dissipated energy must necessarily follow power law with exponent exactly equal to $\epsilon = 1.0$. It was also understood that the reason behind this value of the exponent is the marginal stability of the system [157, 161, 162]. Note that the same arguments have been used to rationalize the observation of the same exponent $\epsilon = 1.0$ in numerical studies of quasi-elastic flow of structural glasses (Tyukodi et al., 2019; Ferrero and Jagla, 2019; Shang et al., 2020) [20, 149, 163].

We recall that marginality of structural glasses is usually linked to the fact that they emerge when liquid is brought exactly to its quenching threshold. Similar type of marginality is also exhibited by jammed states of granular matter that are also expected to be at the threshold between liquid and solid behavior. In both cases 'rigidity' results from the fact that dynamics has just subsided which explains why the ensuing stability is only marginal. In such states the closeness to unstable modes makes the mechanical response inherently intermittent, effectively mixing dynamics and statics. In the same spirit, the recovery of the exponent $\epsilon = 1.0$ in our numerical experiments may be suggesting that our 'amorphous crystal' is posited exactly at the threshold of mechanical stability. This is, in fact, not surprising since the stabilization in the 'amorphous crystal' state was achieved exactly at the end of the first system-size avalanche. Apparently, the associated fast time dynamics brought into the system exactly the level of self-generated disorder needed to transform a stable (elastic) state into a marginally stable (glassy) state. Since we observe the same value of exponent $\epsilon = 1.0$ in both pre- and post-yield regimes, the achieved marginality is not even affected by the catastrophic yielding event. This may mean that the brittle yielding transition does not compromise the global organization of the emerged energy landscape while obviously affecting the reachability of its different subdomains.

In contrast to the exponent ϵ , the cutoff parameter λ differs drastically between the pre-yielding regime, where $\lambda = 0.0230$, and post-yielding regime, where $\lambda = 0.0076$, even though in both cases we deal with the same system size. The approximately three-fold decrease in λ after yielding indicates that large energy drops become substantially more probable in the post-yield regime. This is a clear evidence of an emergent collective behavior which manifests itself in a much broader extent of the post-yield avalanches. Effectively we observe in the post-yield regime an evidence of a divergence of the characteristic length

$$l \sim 1/\lambda. \quad (35)$$

This conjecture is supported by the observation that shear bands in the post-yield regime traverse the whole computational domain reaching the size of the system. This is in stark contrast to unequivocally subextensive localized rearrangements taking place in the pre-yield regime. It is appropriate to evoke here that the divergence of the correlation length is one of the major indicators that the system has reached a critical state. The physical nature of the associated generic (extended) criticality is presently actively debated [164–168].

Finally we mention that our results clearly indicate that the values of parameters ϵ and λ reflect different aspects of the self-organization processes taking place during plastic deformation. The persistence of ϵ over the yielding transition and the variability of λ raises important question about the physical nature of the associated quasi-abrupt emergence of complex long range correlations which ultimately transforms crystallinity into glassiness.[169–171].

V. CONCLUSIONS

In this paper we performed a series of numerical experiments aimed at deeper understanding of the phenomenon of plastic yield in highly idealized 2D crystals in the limit of zero strain rate and negligible thermal effects. To capture the intricate underlying dislocational dynamics we used a relatively novel theoretical and computational tool: a mesoscopic tensorial model (MTM) of crystal plasticity representing a conceptual trade-off between continuum and atomic descriptions.

The MTM is built on the fundamental assumption that mesoscale material elements are exposed to an effective globally periodic energy landscape accounting for lattice-invariant shears. From the perspective of such Landau-type continuum theory with an infinite number of equivalent energy wells, plastically deformed crystal emerges as a multi-phase mixture of equivalent phases. Plastic yield can then be interpreted as an escape from the reference energy well, and plastic mechanisms can be linked to low-energy valleys in the energy landscape. Despite its conservative appearance, the MTM approach is fundamentally dissipative with dry friction dissipation emerging as an averaged description of an overdamped athermal dynamics in a rugged energy landscape. As we explain in the paper, the internal cutoff length scale is brought into the MTM by the assumption that a mesoscopic element (cluster of atoms) deforms homogeneously with inhomogeneity emerging only at supra-mesosopic scales.

In comparison with other mesoscopic theories, the MTM approach has the advantage of describing in a self-consistent manner various topological changes in dislocation configurations such including nucleation, annihilation and interaction with obstacles, while relying only on the information about the inter-atomic potential. In addition to the overall description of plastic yield at the level of stress-strain relations, the MTM approach can capture more subtle effects which depend crucially on the interplay between short and long range interactions among dislocations. An important effect of this type, which was for us of particular interest, is the emergence of intermittent dislocational avalanches.

While it is known that quasi-brittle yielding is a salient feature of well-annealed glassy materials, here we used the MTM approach to show that plastic yielding in model pristine crystals leads to a remarkably similar phenomenology. More specifically, our numerical experiments revealed that plastic yielding in somewhat stylized ideal crystals can be represented as a two stage process.

During the first-stage-yielding massive dislocation nucleation results in the formation of an effectively glassy state. More precisely, a system size avalanche, representing nonlinear elastic instability, a fast time self-organization phenomenon combining nucleation and the proliferation of defects which effectively transform a pristine crystal into what we call an 'amorphous crystal'.

We note that the emergence of such quasi-amorphous state after a massive dislocation nucleation event is not completely surprising. Thus, the mechanical response of conventional poly-crystalline solids is usually qualified as isotropic and is treated by continuum plasticity theory similar to how it treats other amorphous solids. What we see is that at least some features of amorphous response may be acquired by pristine crystals even before the polycrystalline texture develops. In other words, the necessary level of annealed disorder may result from dislocation self-organization at an almost molecular level.

The second-stage-yielding is preceded by pseudo-elastic deformation of a quasi-amorphous solid associated with only minimal dislocational activity. The activity of defects remains localized as in the case of 'caged' rearrangements in amorphous solids. This nominally-elastic stage of plastic deformation which is sometimes interpreted as 'micro-plasticity', ends abruptly with a catastrophic phenomenon closely resembling quasi-brittle yield in well-annealed glasses. We therefore interpret this system spanning event as a transition to the actual plastic yield in a pseudo-amorphous system. The subsequent plastic flow takes the form of intermittent development of a grain texture at almost constant stress. It involves local and global rearrangements of the dislocational structure which takes the form of an evolving system of shear bands.

One of our main findings is that in both pre- and post (second) yield regimes, plastic flow proceeds by local instabilities at the level of single elastic elements which are organized in hierarchical way due to the presence of long range elastic interactions. This self-organization is behind dislocational avalanches which takes place at constant applied strain but which result in stress or energy drops exhibiting power law statistics. Similarly distributed energy and stress drops have been long known to be a signature of amorphous plasticity. The use of the MTM approach allowed us to discover that even the exponents, characterizing power law distributed avalanches, are similar in some model systems representing amorphous and crystalline plasticity. In particular, in full agreement to what is known about plasticity of glassy systems, here we also found that both pre- and post (second) yield avalanches exhibit power law statistics with matching exponents. This supports our idea of an 'amorphous crystal' which is characterized by the marginal stability of the associated defect configuration indicative of a hierarchical structure of the underlying energy landscape.

More work would be needed to convert our initial insights, revealing various peculiar aspects of the behavior of an over simplified prototypical system, into a deep physical understanding of the phenomenon of plastic yield in realistic 3D crystals with, say HCP, FCC and BCC symmetry. For instance, while the developed model deals only with quasi-statically loaded, low dimensional defect free model crystals at zero temperature, it effectively underplays such important physical effects of realistic crystal plasticity as dislocation climb, cross slip, dislocation pinning and forest hardening, to mention just a few. Most importantly, only a 3D model will distinguish adequately edge, screw, and mixed dislocations, and will be suitable to handle topologically nontrivial dislocation entanglements. The 3D model addressing crystallographically nontrivial crystal structures will also allow one to study the effects of the orientation of the samples in the loading device which are known to affect strongly the complexity of the emerging dislocation patterns. To account adequately for all these important physical effects, one would also need to quantitatively calibrate the 3D version of the MTM by a systematic comparing with parallel MD simulations and developing a rigorous conceptual bridge between the notions used to describe atomic and mesoscopic scales.

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